

Assignment: Windows Server – DNS, DHCP, File Sharing & Print Server

1. Configure DNS Server and Create a Forward Lookup Zone

Steps:

1. Open **Server Manager**.
2. Click **Manage → Add Roles and Features**.
3. Select **DNS Server** and install the role.
4. After installation, open **Server Manager → Tools → DNS**.
5. Expand the **Server Name**.
6. Right-click **Forward Lookup Zones** and select **New Zone**.
7. Select **Primary Zone** and click **Next**.
8. Enter the Zone Name as:

myapp.local

9. Click **Next** and complete the wizard.
10. Click **Finish**.
11. Right-click the **myapp.local** zone and select **New Host (A or AAAA)**.
12. Enter the following details:

Name: www

IP Address: 192.168.1.50

13. Click **Add Host**.
14. Click **Done**.

Result:

The **myapp.local** Forward Lookup Zone is created successfully, and the following A record is configured:

www.myapp.local → 192.168.1.50

2. Configure DHCP Server and Create a Scope

Steps:

1. Open **Server Manager**.
2. Click **Manage → Add Roles and Features**.
3. Select and install the **DHCP Server** role.
4. Complete the **DHCP Post-Install Configuration**.
5. Open **Server Manager → Tools → DHCP**.
6. Expand the **Server Name**.
7. Right-click **IPv4** and select **New Scope**.
8. Enter the Scope Name as:

StudentScope

9. Configure the following IP address range:

Start IP Address: 192.168.1.100

End IP Address: 192.168.1.120

10. Set the Subnet Mask as:

255.255.255.0

11. Set the **Lease Duration** to:

8 Hours

12. Configure the **Default Gateway** if required.
13. Continue through the remaining options.
14. Select **Yes, I want to activate this scope now**.
15. Click **Finish**.

Result:

The **StudentScope** DHCP scope is successfully configured and assigns IP addresses from:

192.168.1.100 – 192.168.1.120

with a lease duration of **8 hours**.

3. Share a Folder Named "MusicLibrary"

Steps:

1. Create a folder named **MusicLibrary**.

Example: C:\MusicLibrary

2. Right-click the **MusicLibrary** folder and select **Properties**.
3. Open the **Sharing** tab.
4. Click **Advanced Sharing**.
5. Check **Share this folder**.
6. Set the Share Name as:

MusicLibrary

7. Click **Permissions**.
8. Remove the **Everyone** group.
9. Click **Add** and select the **Students** group.
10. Allow the following Share permissions:
 - Read
 - Change
11. Click **Apply** and then click **OK**.
12. Open the **Security** tab.
13. Add the **Students** group.
14. Grant the following NTFS permissions:
 - Read
 - Write
15. Click **Apply** and then click **OK**.

Result:

The **MusicLibrary** folder is successfully shared. Only users who are members of the **Students** group can read and write files in the shared folder.

4. Install Print Server Role and Share a Printer

Steps:

1. Open **Server Manager**.
2. Click **Manage** → **Add Roles and Features**.
3. Select and install **Print and Document Services**.
4. During the installation, select **Print Server**.
5. Complete the installation.
6. Open **Settings** → **Bluetooth & devices** → **Printers & scanners** or **Devices and Printers**.
7. Click **Add Printer**.
8. Select **Microsoft XPS Document Writer** if a physical printer is not available.
9. Complete the printer installation.
10. Right-click the printer and select **Printer Properties**.
11. Open the **Sharing** tab.
12. Check **Share this printer**.
13. Enter the Share Name as:

CampusPrinter

14. Click **Apply**.
15. Click **OK**.

Result:

The printer is successfully shared with the name **CampusPrinter** and is available for network users.

5. Troubleshooting: Client PC Not Receiving an IP Address

Cause 1: DHCP Server Service is Stopped

Test:

1. Open **Services** by running:

services.msc

2. Locate **DHCP Server**.
3. Check whether the DHCP Server service is running.

Fix:

1. If the service is stopped, right-click **DHCP Server**.
2. Select **Start**.
3. Restart the client computer.
4. Open **Command Prompt** and run:

```
ipconfig /release
```

```
ipconfig /renew
```

Result:

The DHCP Server service is running, and the client should be able to request an IP address from the DHCP server.

Cause 2: DHCP Scope is Inactive or No IP Addresses are Available

Test:

1. Open the **DHCP Console**.
2. Expand the server name.
3. Expand **IPv4**.
4. Check whether the required DHCP scope is **Active**.
5. Check whether free IP addresses are available in the scope.

Fix:

If the scope is inactive:

1. Right-click the scope.
2. Select **Activate**.

If the DHCP address pool is exhausted:

1. Expand the available IP address range if possible.
2. Remove unnecessary or expired leases.
3. Check the scope again for available IP addresses.

Result:

The DHCP scope is active and has available IP addresses for clients.

Cause 3: Client Cannot Communicate with the DHCP Server

Test:

1. Check the physical network cable if a physical computer is being used.
2. If using a virtual machine, check the VM's **Network Adapter** settings.
3. Verify that the client and DHCP server are connected to the same network.
4. Test connectivity using:

```
ping <DHCP_Server_IP>
```

Example:

```
ping 192.168.1.10
```

Fix:

1. Connect the client to the correct network.
2. Ensure that the network adapter is enabled.
3. Restart the network adapter.
4. If using a virtual machine, reconnect or restart the virtual network adapter.
5. Run the following commands again:

```
ipconfig /release
```

```
ipconfig /renew
```

Result:

The client is connected to the correct network and can communicate with the DHCP server.

